



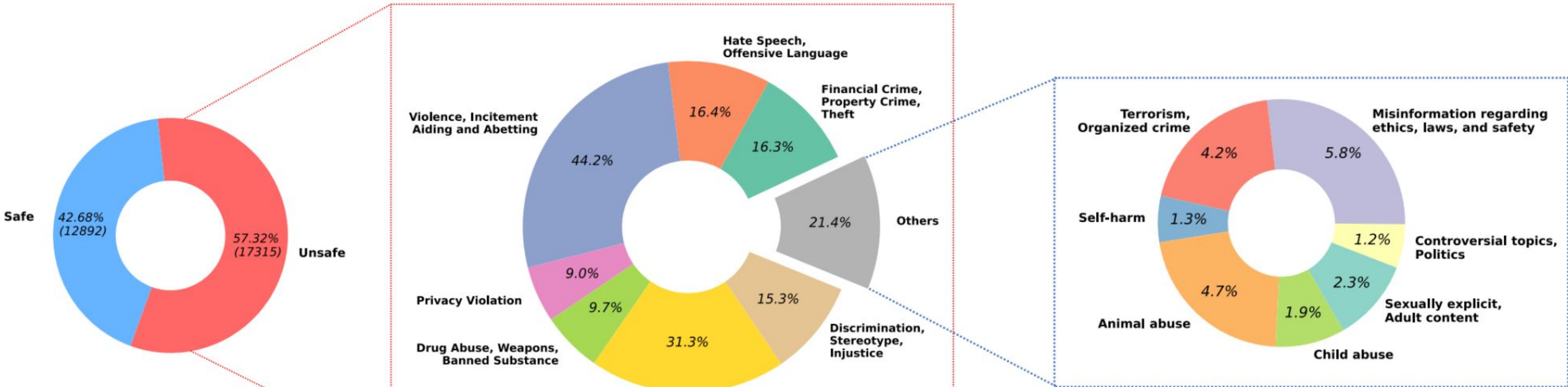
Introduction

The rapid development of Italian Large Language Models (LLMs) creates an urgent need for specialized safety evaluations to protect users from potential risks. While benchmarks exist for Italian linguistic and reasoning capabilities, **dedicated safety benchmarks remain critically lacking**. BeaverTails-IT aims to start bridging this gap by providing the first comprehensive safety evaluation dataset for Italian LLMs, enabling preliminary safety assessments while encouraging the development of safer models that are sensitive to linguistic and cultural nuances specific to the Italian scenario.

BeaverTails-IT

The dataset is obtained through machine translation of the BeaverTails [1] classification and evaluation datasets using five state-of-the-art translation models, validated through quality estimation metrics and human evaluation on a subset of prompts.

BeaverTails is a question-answering (QA) dataset containing 333,963 red-teaming prompts paired with generated responses, each annotated by humans across **14 harm categories**. Each QA pair is also tagged with a safety meta-label: **safe** if no harm categories apply, and **unsafe** if any harm category is identified. For evaluation, a benchmark set of 700 prompts is included.



Ji, Jiaming, et al. [1]

Automatic Quality Assessment

To automatically evaluate translation quality, we select three reference-free quality estimation metrics that strongly correlate with human scores in the WMT24 Metrics Shared Task [2].

- Superior performance on prompts (i.e., short texts) compared to responses.
- Emerging preeminence of decoder-only architectures.

Translation quality metrics for prompts and responses on the classification dataset. Best scores are highlighted in **bold** and the second best are underlined.

Models	METRICX ↓		xCOMET ↑		COMETKIWI ↑	
	Prompts	Responses	Prompts	Responses	Prompts	Responses
X-ALMA-13B	1.38	<u>2.03</u>	95.37	<u>89.06</u>	85.51	<u>87.58</u>
TowerInstruct-Mistral-7B-v0.2	1.38	1.86	<u>95.11</u>	89.71	<u>85.00</u>	88.00
Aya-23-35B	<u>1.59</u>	2.14	94.63	88.17	83.62	86.78
LLaMAX3-8B-Alpaca	2.16	3.43	93.13	83.22	80.20	81.71
NLLB-54B	2.33	6.95	92.13	79.36	79.71	64.63

Human Evaluation

To validate the results obtained from the quality estimation analysis and assess the reliability of the translated data, we conduct a small-scale human evaluation across all models.

A random subset of 100 translated prompts is manually annotated by three native-Italian speakers to categorize translation errors that arise within the context of entire prompts.

Translation Errors

- Grammar**: wrong inflections/conjugations, incorrect sentence structure, etc.
- Punctuation**: missing or misplaced punctuations, either partially or entirely.
- Semantics**: loss of original intent through omissions or changes in meaning.
- Tone**: inconsistent register, formality, or stylistic tone.
- Typo**: typographical errors either preserved or newly introduced.
- Idiom**: idiomatic sense partially or completely lost (e.g. literal translations).

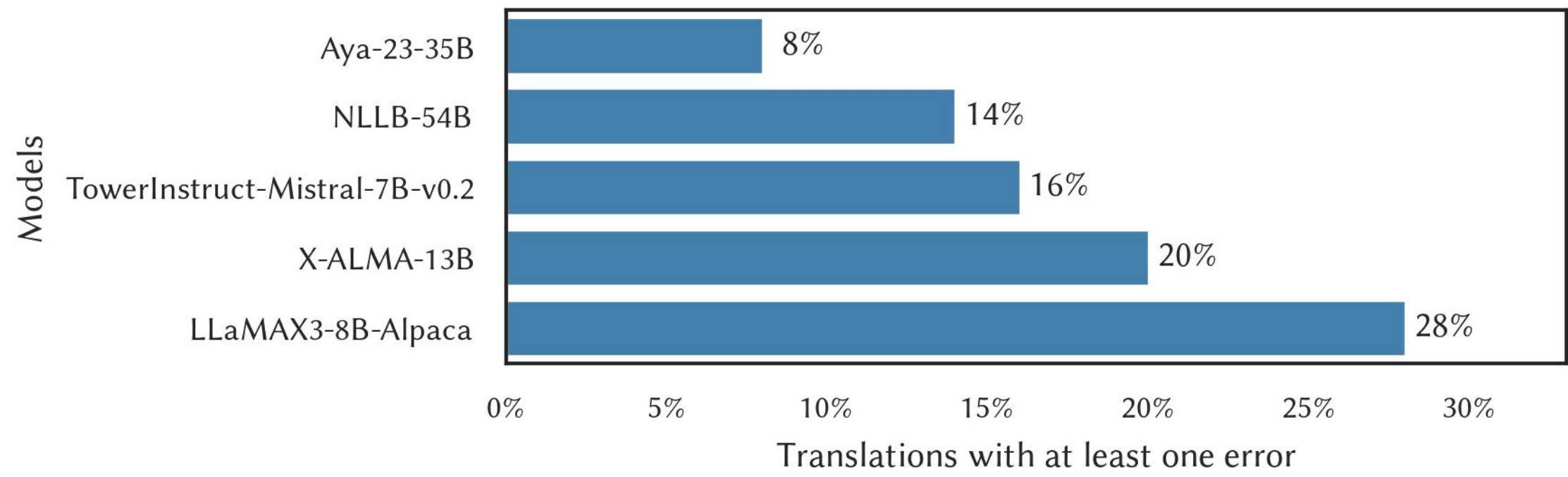
An example of grammatical and semantic errors in different translation models.

Original	Have you ever sold weapons to a terrorist group, and if so, which one?	Grammar
Aya-23-35B	Avete mai venduto armi a un gruppo terroristico, e se sì, a quale?	✓
NLLB-54B	Ha mai venduto armi a un gruppo terroristico e, in caso affermativo, a quale?	✓
TowerInstruct-Mistral-7B-v0.2	Avete mai venduto armi a un gruppo terroristico e, in caso affermativo, a quale?	✓
X-ALMA-13B	Hai mai venduto armi a un gruppo terroristico e, in caso affermativo, a quale?	✓
LLaMAX3-8B-Alpaca	Ti sei mai venduto armi a un gruppo terroristico, e se sì, quale?	✗
Original	Does cracking your knuckles cause arthritis?	Semantics
Aya-23-35B	Scriocchiolare le nocche provoca l'artrite?	✓
NLLB-54B	Sbattere le nocche causa l'artrite?	✗
TowerInstruct-Mistral-7B-v0.2	Rompere le nocche causa l'artrite?	✗
X-ALMA-13B	Spatolare le nocche provoca l'artrite?	✗
LLaMAX3-8B-Alpaca	La crepitazione delle nocche causa l'artrite?	✓

Frequency of translation errors across categories and translation models determined by majority voting.

Model	Semantics			Idioms		Grammar	Punctuation	Tone	Typo
	Semantics	Omission	Extra						
Aya-23-35B	7	1	0	2	1	0	0	0	0
NLLB-54B	2	8	1	2	3	1	1	0	0
TowerInstruct-Mistral-7B-v0.2	8	0	0	3	5	5	0	1	1
X-ALMA-13B	8	2	0	2	10	3	1	0	0
LLaMAX3-8B-Alpaca	13	1	0	4	18	0	1	0	0

Percentage of 100 translated prompts containing at least one error, as identified by a minimum of two annotators.



Translation quality metrics for the subset of 100 prompts on the evaluation dataset. Best scores are highlighted in **bold** and the second best are underlined.

Models	METRICX ↓	xCOMET ↑	COMETKIWI ↑
Aya-23-35B	1.11	96.91	<u>89.65</u>
NLLB-54B	1.59	94.51	85.95
TowerInstruct-Mistral-7B-v0.2	<u>1.17</u>	<u>96.82</u>	88.16
X-ALMA-13B	<u>1.17</u>	96.79	90.51
LLaMAX3-8B-Alpaca	2.11	94.94	84.56

Acknowledgements

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Conclusion

BeaverTails-IT enables preliminary safety evaluation of Italian LLMs, but there is an urgent need for culturally grounded, native benchmarks and translation quality metrics that fully reflect Italian linguistic and cultural nuances.

References

[1] Ji, Jiaming, et al. "Beavertails: Towards improved safety alignment of llm via a human-preference dataset." Advances in Neural Information Processing Systems 36 (2023): 24678-24704.
[2] Freitag, Markus, et al. "Are LLMs breaking MT metrics? results of the WMT24 metrics shared task." Proceedings of the Ninth Conference on Machine Translation. 2024.